JING LIU

PhD Researcher — LLMs · Interpretability · RL & Evaluation

CONTACT

ecnucrystal@gmail.com

@Jing-L97

@Jing Liu

+33 758 828 907

SUMMARY

PhD researcher and **Marie Skłodowska-Curie Fellow** working on large language models: **mechanistic interpretability**, **RL fine-tuning** (PPO/DPO/GRPO), **retrieval augmentation**, and **automatic evaluation**. I build data-efficient training and evaluation pipelines and turn them into reusable benchmarks and baselines adopted by the community (BabyLM 2025). First-author work at NeurIPS, ICML, COLM, and CogSci. Area Chair at EMNLP; co-organizer of BabyLM 2025/2026.

SKILLS

Languages
Python, R, MATLAB, Java, Prolog, JavaScript, HTML, Bash

ML & DL Frameworks
PyTorch, TensorFlow, Fairseq, Hugging Face TRL, vLLM

LLMs & Alignment
RLHF (PPO, DPO, GRPO), reward modeling, RAG, LLM-as-a-judge evaluation

Interpretability
Logit attribution (LogitLens), circuit tracing, sparse autoencoders (SAE)

Speech & Audio
Self-supervised speech models (HuBERT, wav2vec 2.0), speaker diarization (pyannote), forced alignment, prosody (openSMILE), Praat, ELAN

EDUCATION

Ph. D. - NLP & Cognitive Modeling — **2023 - ongoing**
École Normale Supérieure, Paris, France
Thesis: Data-Efficient Spoken Language Modeling (Advisors: Emmanuel Dupoux, Najoung Kim, David Hawarth)

Adv. Master - Artificial Intelligence — **2022 - 2023**
KU Leuven, Leuven (Belgium)
Thesis: Computational Modeling of Early Word Acquisition.

Research Master - Linguistics & Comm. Sciences (Distinction) — **2020 - 2022**
Radboud University, Nijmegen (the Netherlands)
Thesis: Predicting Backchannels in Multimodal Child-Caregiver Conversation (Advisors: Abdellah Fourtassi, Asli Özyürek)

BA in English education & BSc in Psychology — **2015 - 2019**
East China Normal University, Shanghai, China
Thesis: Mnemonic Effects in Idiom Learning. (Advisor: Yanning Yang)

WORK

Applied Scientist Intern — **2026**
Amazon (AICE Team), Berlin, Germany
Building data-driven automation for large-scale Retail/IT service operations with time-series anomaly detection, NLP, and GenAI with LLM-as-a-judge evaluation.

Research Intern — **2022**
Inria, Paris, France
Built multimodal models to predict backchannels and turn-taking in a child–caregiver conversation corpus (audio, video, transcripts; prosodic, gaze, and lexical features), showing adult-like turn coordination emerges by middle childhood. Published at ICMI 2022 and CogSci 2023.

English Teacher, Senior High School — **2019 - 2020**
No. 2 Hohhot Senior High School, Inner Mongolia, China
Taught English and introductory psycholinguistics to senior high-school students; served as homeroom teacher, owning daily management, and academic guidance for a class of 50 students.

PUBLICATIONS

Peer-Reviewed (Conferences & Workshops)

Methods
Benchmark & evaluation design, ablation studies, statistical analysis, data-efficient / curriculum training

Infra & Tooling
Git, Linux, Weights & Biases, LaTeX

ACHIEVEMENTS

Fellowships

Marie-Curie PhD Fellow (2023)

Radboud Honors Academy

National Research Training Program

Scholarships & Awards

Merit Award, Pudong Global Entrepreneurship Competition (2026)

NeurIPS Travel Fund (2025)

INTERSPEECH Student Grant

Radboud Scholarship

2nd Prize – National Teaching Competition

ECNU Guanghua Scholarship

ECNU Scholarship

- **Liu, J.**, & Shukla, A. (2026). Compression at a Cost: Interpreting Information Bottlenecks in Safety-Aligned Reward Models. Mechanistic Interpretability Workshop, ICML, Seoul, South Korea.
- **Liu, J.**, & Kim, N. (2026). Does Episodic Memory Help Close the Lexical Frequency Gap in Sensitivity to Syntactic Contrasts? A Test Using Retrieval-Augmented Language Models. COLM, San Francisco, USA.
- **Liu, J.**, Li, Y., & Wang, H. (2025). Distributed Specialization: Rare-Token Neurons in Large Language Models. Mechanistic Interpretability Workshop, NeurIPS, San Diego, USA.
- **Liu, J.** (2025). No Clustering, No Routing: How Transformers Actually Process Rare Tokens. UniReps Workshop, NeurIPS, San Diego, USA.
- **Liu, J.**, Wang, H., & Li, Y. (2025). Emergent Specialization: Rare Token Neurons in Language Models. High-dimensional Learning Dynamics Workshop, ICML, Vancouver, Canada.
- **Liu, J.**, & Fourtassi, A. (2025). Benchmarking LLMs for Mimicking Child-Caregiver Language in Interaction. Cognitive Science Society, San Francisco, USA.
- Charpentier, L., Choshen, L., Cotterell, R., Gul, M. O., Hu, M., Jumelet, J., Linzen, T., **Liu, J.**, Mueller, A., Ross, C., Shah, R. S., Warstadt, A., Wilcox, E., & Williams, A. (2025). Findings of the Third BabyLM Challenge. Proceedings of the Third BabyLM Workshop, ACL.
- Agrawal, A., **Liu, J.**, Bodur, K., Favre, B., & Fourtassi, A. (2023). Development of Multimodal Turn Coordination in Conversations: Evidence for Adult-like Behavior in Middle Childhood. Cognitive Science Society, Sydney, Australia.
- **Liu, J.**, Nikolaus, M., Bodur, K., & Fourtassi, A. (2022). Predicting Backchannel Signaling in Multimodal Child-Caregiver Conversation. WoCBU Workshop, ICMI 2022, Bangalore, India.

Preprints & Under Review

- **Liu, J.**, Schweitzer, S., & Fourtassi, A. (2026). Which Forms of Caregiver Feedback Support Grammar Learning? A Reinforcement-Learning Study of Child-Like Language Models. arXiv.
- **Liu, J.**, & Xie, S. (2026). Is Curriculum Learning Based on Pedagogical Child-Directed Speech Effective for Language Model Training? arXiv.
- **Liu, J.**, & Dupoux, E. (2026). On Comparing Developmental Curves in Children and AI Simulations. Annual Review of Developmental Psychology (under review).
- **Liu, J.** (2025). Layer Specialization Underlying Compositional Reasoning in Transformers. arXiv.
- **Liu, J.**, Lavechin, M., Hamilakis, N., & Dupoux, E. (2025). CORPUS-CDI: Comparing Children and AI Models' Learning Rate of Expressive Vocabulary. arXiv.
- **Liu, J.** (2025). Circuit-Aware Reward Training: A Mechanistic Framework for Longtail Robustness in RLHF. arXiv.
- **Liu, J.**, Lavechin, M., Hamilakis, N., & Dupoux, E. (2025). Machine CDI: A Lexical Benchmark for Expressive Vocabulary (submitted).
- Charpentier, L., Choshen, L., Cotterell, R., Gul, M. O., Hu, M., Jumelet, J., Linzen, T., **Liu, J.**, Mueller, A., Ross, C., Shah, R. S., Warstadt, A., Wilcox, E., & Williams, A. (2025). BabyLM Turns 3: Call for Papers for the 2025 BabyLM Workshop. arXiv.

SELECTED PROJECTS

LLM Alignment & RL Fine-Tuning

Communicative-Feedback Alignment (BabyLM 2025 baseline) — 2024
Official baseline, BabyLM Challenge 2025

Implemented PPO-based RLHF to align child-like LM generations with caregiver feedback signals; released as an official baseline adopted by BabyLM 2025 participants.

Multi-Reward Feedback Alignment — 2026
Submitted to EMNLP 2026

Built reward aggregation with automatic weighting to align LM generations across multiple caregiver-feedback signals.

Mechanistic Interpretability

MI-Reward: Interpreting Information Bottlenecks in Safety-Aligned Reward Models — 2025-26
Mechanistic interpretability Work shop ICML 2026

Built an SAE-based reward-modeling pipeline to identify spurious features that drive reward hacking in RLHF.

Emergent Specialization in LLMs — 2025
NeurIPS MI Workshop; ICML HiLD Workshop, 2025

Characterized rare-token neuron specialization in transformers via logit attribution and sparse autoencoders, showing distributed (not modular) processing of rare tokens.

Compositional Reasoning in Grokked Transformers — 2025

Analyzed compositional in-context reasoning using circuit tracing and a parameter-efficient attention design.

Evaluation & Benchmarks

Machine CDI: Lexical Benchmark for LLMs — 2023-24
Submitted

Constructed a lexical benchmark grounding LLM expressive-vocabulary evaluation in developmental (CDI) norms, enabling direct child-vs-model comparison.

LLM Conversational Agents & Evaluation Suite — 2024
CogSci 2025; metrics, BabyLM 2025

Built LLM-based agents simulating child–adult dialogue and a multi-level linguistic evaluation suite; contributed evaluation metrics to the BabyLM 2025 Challenge.

Retrieval-Augmented LMs for Rare-Tokens — 2024-25
COLM 2026

Studied which retrieval factors let RAG close the lexical-frequency gap in syntactic sensitivity, using controlled retrieval-augmented evaluation.

Speech & Multimodal Modeling

Speech LM Evaluation for Lexical Discrimination — 2026
Ongoing

Evaluated self-supervised speech representations on lexical-discrimination tasks for spoken language models.

Multimodal Conversational-Behavior Prediction 2022-23
ICMI 2022; CogSci 2023

Built multimodal models to predict backchanneling and turn-taking from audio, video, and transcripts.

SERVICE & LEADERSHIP

- **Area Chair**, ACL Rolling Review (EMNLP), 2026
- **Co-organizer**, BabyLM Challenge 2025; BabyLM-Speech 2026; Computational Developmental Linguistics, ACL 2026
- **Invited Lecturer**, Boston University, Spring 2024
- **Reviewer**: ICML & NeurIPS Mechanistic Interpretability / UniReps Workshops (2025–26), ACL 2025, BabyLM (ACL/EMNLP), *Open Mind* journal
- **Organizer**, Weekly NLP Reading Group, 2024–present
- **Teaching**: Psycholinguistics & Comprehensive English (2019–20); TA, English Phonetics (2017–18)